

Week 3: Co-movements of financial data

Multiple asset returns?

- ▶ When building portfolios, the investor wishes to know how assets move together.
- ▶ These features tell us about the potential risk in investing in an asset.
- ▶ How should we measure these?

Joint Probability

Marginal Probability

Conditional Probability

Joint Probability Distribution

Let X and Y be discrete random variables with sample spaces S_X and S_Y , respectively.

The likelihood that X and Y take values in the joint sample space $S_X \times S_Y$ is determined by the *joint probability distribution*

$$p(x, y) = \Pr(X = x, Y = y).$$

The function $p(x, y)$ satisfies:

$$\sum_{x \in S_X} \sum_{y \in S_Y} \Pr(X = x, Y = y) = 1.$$

A simple example

Let E denote the performance of the economy next quarter.

For simplicity, say $E = 1$ if the economy is expanding and $E = 0$ if the economy is contracting.

E is a discrete random variable as we are able to list all the possible outcomes.

The sample space for E is $S_E = \{0, 1\}$.

Let S represent my sales for the next quarter, and assume the sample space for S is $S_S = \{0, 1, 2, 3\}$.

Table 1 (see the next slide) illustrates the joint distribution for S and E .

Table 1: Bivariate Distribution

s	0	1	$\Pr(S)$
0	$1/8$	0	$1/8$
1	$2/8$	$1/8$	$3/8$
2	$1/8$	$2/8$	$3/8$
3	0	$1/8$	$1/8$
$\Pr(E)$	$4/8$	$4/8$	1

Joint Probability Distribution

Notice that the total of all the entries in the table adds up to one.

$$\sum_{s \in S_S} \sum_{e \in S_E} \Pr(S = s, E = e) = 1.$$

The joint probability distribution tells us the probability of S and E occurring together.

What if we only want to know about the probability of S occurring, or the probability of E occurring?

Marginal probability

The probability $\Pr(X = x)$ is called the *marginal probability* of X and is given by:

$$\Pr(X = x) = \sum_{y \in S_Y} \Pr(X = x, Y = y).$$

Similarly, the *marginal probability* of $Y = y$ is given by:

$$\Pr(Y = y) = \sum_{x \in S_X} \Pr(X = x, Y = y).$$

Find $\Pr(S = 0)$ and $\Pr(E = 1)$ from the joint distribution in Table 1.

What is $\Pr(S = 0)$ regardless of the value of E ?

$$\begin{aligned}\Pr(S = 0) &= \Pr(S = 0, E = 0) + \Pr(S = 0, E = 1) \\ &= \frac{1}{8} + 0 \\ &= \frac{1}{8}\end{aligned}$$

Notice that this probability is equal to the horizontal (row) sum of the probabilities in Table 1 at $S = 0$.

Marginal Distributions

Similarly, we can find $\Pr(E = 1)$ as follows:

$$\begin{aligned}\Pr(E = 1) &= \Pr(S = 0, E = 1) + \Pr(S = 1, E = 1) + \\ &\quad \Pr(S = 2, E = 1) + \Pr(S = 3, E = 1) \\ &= 0 + \frac{1}{8} + \frac{2}{8} + \frac{1}{8} \\ &= \frac{4}{8}\end{aligned}$$

This probability is the vertical (column) sum of the probabilities in Table 1 at $E = 1$.

Marginal Distributions

Why we call *marginals* probability “marginals” ?

The table represents the joint and in the margins, we get the marginals.

Conditional Distributions

The conditional probability that $X = x$ given that $Y = y$ (provided $\Pr(Y = y) \neq 0$) is,

$$p(x|y) = \Pr(X = x|Y = y) = \frac{\Pr(X = x, Y = y)}{\Pr(Y = y)}$$

The conditional probability that $Y = y$ given that $X = x$ (provided $\Pr(X = x) \neq 0$) is,

$$p(y|x) = \Pr(Y = y|X = x) = \frac{\Pr(X = x, Y = y)}{\Pr(X = x)}$$

Conditional Distributions

What is the probability that $S = 0$ given that we know $E = 0$?

To find this probability, we compute the conditional probability:

$$\Pr(S = 0|E = 0) = \frac{\Pr(S = 0, E = 0)}{\Pr(E = 0)} = \frac{1/8}{4/8} = \frac{1}{4}.$$

The notation $\Pr(S = 0|E = 0)$ is read as “the probability that $S = 0$ given that $E = 0$ ”.

Conditional Distributions

Notice that

$$\Pr(S = 0|E = 0) = \frac{1}{4} > \Pr(S = 0) = \frac{1}{8}.$$

Hence, knowledge that $E = 0$ increases the likelihood that $S = 0$.

Clearly, S *depends on* E .

Table 2: Conditional distributions

s	$\Pr(S E = 0)$	s	$\Pr(S E = 1)$
0	$2/8$	0	0
1	$4/8$	1	$2/8$
2	$2/8$	2	$4/8$
3	0	3	$2/8$

The first column is the conditional distribution of S given $E = 0$.

The second column is the conditional distribution of S given $E = 1$.

We read: the probability of Sales of 3 ($S = 3$) given (or conditional on) the economy is growing ($E = 1$) is $2/8$.

Discrete random variables

In general the notation is...

- ▶ $\Pr(Y = y, X = x)$ is the **joint probability** of the random variable Y equal y **AND** the random variable X equal x .
- ▶ $\Pr(Y = y)$ and $\Pr(X = x)$ are the **marginal probabilities** of $Y = y$ and $X = x$.
- ▶ $\Pr(Y = y \mid X = x)$ is the **conditional probability** of the random variable Y takes the value y **GIVEN** that X equals x .

Independence

Intuitively, X is independent of Y if knowledge about Y does not influence the likelihood that $X = x$.

We represent this intuition formally as follows.

Let X and Y be discrete random variables with sample spaces S_X and S_Y , respectively. X and Y are independent random variables if and only if:

$$\Pr(X = x|Y = y) = \Pr(X = x), \quad \text{for all } x \in S_X, y \in S_Y$$

Independence

For the data in Tables 1 and 2, we know that

$$\Pr(S = 0|E = 0) = \frac{1}{4} \neq \Pr(S = 0) = \frac{1}{8}.$$

Thus S and E are not independent.

Expected Value and Conditional Expectation

Expected Value

For a discrete random variable X with sample space S_X , the expected value is defined as:

$$\mathbb{E}[X] = \sum_{x \in S_X} x \cdot \Pr(X = x).$$

We weight each possible value by how likely they are. . .

This provides us with a measure of centrality of the distribution.

A “good” prediction for X !

Table 1: Bivariate Distribution

s	0	1	$\Pr(S)$
0	$1/8$	0	$1/8$
1	$2/8$	$1/8$	$3/8$
2	$1/8$	$2/8$	$3/8$
3	0	$1/8$	$1/8$
$\Pr(E)$	$4/8$	$4/8$	1

Expected Value

Using the discrete distribution in Table 1, the expected return is computed as:

$$\begin{aligned}\mathbb{E}[S] &= 0 \cdot (1/8) + 1 \cdot (3/8) + 2 \cdot (3/8) + 3 \cdot (1/8) \\ &= 1.5\end{aligned}$$

Conditional Expectation

$\mathbb{E}[X|Y = y]$ is the conditional mean of $X|Y = y$.

For discrete random variables X and Y , the conditional expectations are defined as:

$$\mathbb{E}[X|Y = y] = \sum_{x \in S_X} x \cdot \Pr(X = x|Y = y),$$

$$\mathbb{E}[Y|X = x] = \sum_{y \in S_Y} y \cdot \Pr(Y = y|X = x).$$

Table 2: Conditional distributions (duplicate)

s	$\Pr(S E = 0)$	s	$\Pr(S E = 1)$
0	$2/8$	0	0
1	$4/8$	1	$2/8$
2	$2/8$	2	$4/8$
3	0	3	$2/8$

Conditional Expectation

For the random variables in Table 2 (Conditional Distributions), we have the following conditional moments for S :

$$\mathbb{E}[S|E = 0] = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{4} + 3 \cdot 0 = 1$$

$$\mathbb{E}[S|E = 1] = 0 \cdot 0 + 1 \cdot \frac{1}{4} + 2 \cdot \frac{1}{2} + 3 \cdot \frac{1}{4} = 2$$

Before E is observed, $\mathbb{E}[S|E]$ is a random variable because its value depends on the value of E .

Since $\mathbb{E}[S|E]$ is a random variable, we can consider their mean values.

To illustrate:

$$\begin{aligned} & E[S|E = 0] \times \Pr(E = 0) + E[S|E = 1] \times \Pr(E = 1) \\ &= 1 \times \frac{1}{2} + 2 \times \frac{1}{2} \\ &= 1.5 \\ &= E(S) \end{aligned}$$

This result is known as the **Law of Iterated Expectations** (LIE).

Law of Iterated Expectations

For two random variables X_1 and X_2 (discrete or continuous), the Law of Iterated Expectations (LIE) states that

$$\mathbb{E}[X_1] = \mathbb{E}[\mathbb{E}[X_1|X_2]].$$

In words, the LIE says that the mean of the conditional mean is the unconditional mean.

Intuition: LIE

1. The inner expectation $E[X_1|X_2]$ represents the conditional expectation of X_1 given X_2 . It treats X_2 as known.
2. The outer expectation $E[\cdot]$ then takes the expectation of that conditional expectation over all possible values of X_2 .

Why is it useful?

It simplifies calculations when direct computation of $E[X_1]$ is difficult.
It is widely used in finance and econometrics.

Conditional variance

A very useful conditional moment is the conditional variance.

Definition

The variance of a random variable Y , conditional on X is given by

$$\text{Var}(Y|X) = \mathbb{E} \left[(Y - \mathbb{E}[Y|X])^2 | X \right].$$

Law of Total Variance:

The variance of Y can be equivalently calculated as

$$\text{Var}(Y) = \mathbb{E}[\text{Var}(Y|X)] + \text{Var}(\mathbb{E}[Y|X]).$$

Continuous Multivariate Random Variables

Joint Distributions

Multivariate random variables allow relationship between two or more random quantities to be modelled and studied.

We will use bold face type to indicate a column vector with n components

$$\mathbf{X} = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{pmatrix}$$

Each of the X_i s is a random variable.

Joint Distributions

- ▶ Multivariate random variables can be either discrete or continuous.
- ▶ Discrete multivariate random variables are fairly uncommon in financial econometrics and so we will focus on the continuous case.
- ▶ The joint distribution of multivariate random variables can be easily understood in the bivariate case.
- ▶ Consider the bivariate case: $\mathbf{X} = (X_1, X_2)'$.

Joint Distributions

Let $\mathbf{X} = (X_1, X_2)'$ be continuous random variables defined over the real line.

We characterize the joint probability distribution of X_1 and X_2 using the joint probability density function (pdf) $f(x_1, x_2)$ such that $f(x_1, x_2) \geq 0$ and,

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x_1, x_2) dx_1 dx_2 = 1.$$

Marginal Distribution

Let $\mathbf{X} = (X_1, X_2)'$ with joint pdf $f(x_1, x_2)$.

The marginal pdf of X_1 is given by

$$f_1(x_1) = \int_{-\infty}^{\infty} f(x_1, x_2) dx_2.$$

Likewise, the marginal pdf of X_2 is given by

$$f_2(x_2) = \int_{-\infty}^{\infty} f(x_1, x_2) dx_1.$$

The marginal distribution requires integrating out the variables which are not of interest.

If X_1 and X_2 are jointly normally distributed, then their joint behaviour is completely characterised by the mean vector and covariance matrix:

$$\begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim N \left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \sigma_{11}^2 & \sigma_{12} \\ \sigma_{12} & \sigma_{22}^2 \end{pmatrix} \right)$$

where $\sigma_{12} = \text{Cov}(X_1, X_2)$.

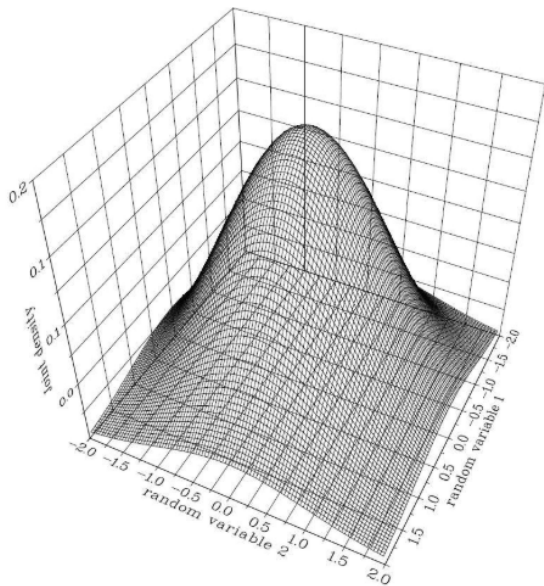
The joint pdf is given by

$$f(x_1, x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} e^{-\frac{1}{2\sqrt{1-\rho^2}} \left[\frac{(x_1-\mu_1)^2}{\sigma_1^2} + \frac{(x_2-\mu_2)^2}{\sigma_2^2} - \frac{2\rho(x_1-\mu_1)(x_2-\mu_2)}{\sigma_1\sigma_2} \right]}$$

where

$$\rho = \frac{\sigma_{12}}{\sigma_1\sigma_2}.$$

Joint density for two standard normally distributed random variables and $\rho = 0.5$.



Conditional Distributions

- ▶ Our goal will often be to understand the behavior of X_1 , conditional on X_2 , taking a specific value $X_2 = x_2$ or within some range of values.
- ▶ Vast majority of financial econometrics is interested in understanding the behavior of a random variable conditional on the observed values of some other random variables.
- ▶ We define the conditional distribution of a random variable, or its moments, by making use of the joint and marginal distributions.
- ▶ Consider $\mathbf{X} = (X_1, X_2)'$.

Conditional Distribution

Let $\mathbf{X} = (X_1, X_2)'$ be a bivariate random variable with joint pdf $f(x_1, x_2)$ and marginal densities $f_1(x_1)$ for X_1 , and $f_2(x_2)$ for X_2 .

The **conditional pdf** of X_1 given $X_2 = x_2$, is

$$f(x_1|x_2) = \frac{f(x_1, x_2)}{f_2(x_2)}.$$

Independence

Let $\mathbf{X} = (X_1, X_2)'$ with joint pdf $f(x_1, x_2)$ and marginal pdfs $f_1(x_1)$, $f_2(x_2)$.

The random variables X_1 and X_2 are independent if and only if

$$f(x_1, x_2) = f_1(x_1)f_2(x_2).$$

This result is extremely useful in practice because it gives us an easy way to compute the joint pdf for two independent random variables: we simply compute the product of the marginal distributions.

Independence

Let X_1 and X_2 be continuous random variables with marginal pdfs $f_1(x_1)$ and $f_2(x_2)$, and joint pdf $f(x_1, x_2)$.

Then X_1 and X_2 are independent if and only if

$$f(x_1|x_2) = \frac{f(x_1, x_2)}{f_2(x_2)} = \frac{f_1(x_1)f_2(x_2)}{f_2(x_2)} = f_1(x_1).$$

That is, if variables are independent, the conditional distribution is nothing but the marginal distribution.